Unit 5.5 Integration

Note Title 2016/05/18

Note:

- Anti-derivate
  (also indefinite integral)
- Definite integral
- Area

Integral types:

I. $\int x^n \cdot dx$

$$\int (f(x))^n \cdot f'(x)\, dx$$

<u>Examples:</u>

$$\int \left(x^4 + \frac{1}{x^4}\right) dx$$

$$\int 2x(x^2+1)\, dx$$

$$\int \left(\sqrt{x} + \frac{1}{\sqrt{x}}\right) dx$$

$$\int 2x(1+x^2)^3\, dx$$

$\int x \sqrt{1+x^2}\, dx$

II. $\int \sin ax \, dx = \frac{-1}{a} \cos ax + c.$

$\int \cos x . (\sin x)^n . dx = \frac{1}{n+1} (\sin x)^{n+1} + c.$

$\int f'(x) \sin(f(x)) dx = -\cos(f(x)) + c$

Examples

$\int \sin \frac{x}{2}\, dx$

$\int \sin x . \cos^3 x . dx$

$\int x^2 . \sin(x^3+1)$

$\int \sin x . \cos x . dx$

$\int \sqrt{\sin 5x} . \cos 5x . dx$

I and II mixed examples

$$\int x^4 (1 + x^5)^6 \, dx$$

$$\int (x+1)(x^2+2x)\, dx$$

$$\int \frac{1}{\sqrt{x}} \cdot \sin\sqrt{x} \cdot dx.$$

$$\int x.\cos x^2.\sqrt{\sin x^2}\, dx.$$

$$\int (x - \sin 5x)^4 (1 - 5\cos 5x)\, dx$$

$$\int \cos^2 4x \cdot \sin 4x \cdot dx$$

III. $\int e^{ax} \, dx = \frac{1}{a} e^{ax} + c$

$$\int f'(x) \cdot e^{f(x)} \, dx = e^{f(x)} + c.$$

Examples

$$\int e^{3t} \, dt$$

$$\int x^2 \cdot e^{x^3+1} \, dx$$

$$\int e^{\tan 3x} \cdot \sec^2 3x \, dx$$

$$\int \sec^2(e^x) \cdot e^x \, dx$$

$$\int \cos x \cdot e^{3\sin x} \, dx = \frac{1}{3} e^{3\sin x} + c.$$

$\underline{IV}$. $\displaystyle\int \frac{f'(x)}{f(x)}\,dx = \ln|f(x)| + C.$

Examples

$$\int \frac{1}{2x+1}\,dx$$

$$\int \frac{\cos x}{1+2\sin x}\,dx$$

$$\int \frac{\cos x}{(1+2\sin x)^3}\,dx$$

$$\int \frac{\sec x\tan x + \cos x}{\sec x + \sin x}\,dx$$

$$\int \frac{dx}{\sqrt{x}\,.\,\ln(\sqrt{x}+1) \times (\sqrt{x}+1)}$$

V. $\int \frac{1}{\sqrt{1-x^2}}\,dx = \arcsin x + c$

or $\sin^{-1} x + c$.

$$\int \frac{f'(x)}{\sqrt{1-(f(x))^2}}\,dx = \arcsin f(x) + c.$$

$$\int \frac{1}{1+x^2}\,dx = \arctan x + c.$$

$$\int \frac{f'(x)}{1+(f(x))^2}\,dx = \arctan f(x) + c.$$

Examples

$$\int \frac{1}{\sqrt{1-x^2}}\,dx$$

$$\int \frac{x}{\sqrt{1-x^2}}\,dx$$

$$\int \frac{1}{1+x^2}\,dx$$

$$\int \frac{x}{1+x^2}\,dx$$

$$\int \frac{x}{(1+x^2)^2}\,dx$$

$$\int \frac{x^2}{1+x^6}\,dx$$

$$\int \frac{e^{2x}}{1-e^{4x}}\,dx$$

$$\int \frac{\sec^2 x}{\sqrt{1-\tan^2 x}}\,dx$$

Two handy forms

$$\int \sqrt{a^2-x^2}\,dx = \arcsin \frac{x}{a} + C$$

$$\int \frac{dx}{a^2+x^2} = \frac{1}{a} \arctan \frac{x}{a} + C.$$

Examples

$$\int \sqrt{9-x^2}\,dx$$

$$\int \frac{1}{4+x^2}\,dx$$

Two "plans".

1. $\int \frac{x^4+1}{x^3}\,dx$

$$\cancel{\frac{x^3}{x^4+1} = \frac{x^3}{x^4}} + \frac{x^3}{1}$$

2. $\int \frac{x^2}{x^2+1}\,dx$

VI $\int a^x \cdot dx = \frac{1}{\ln a} \cdot a^x + c$

$\int f'(x)\, a^{f(x)} \cdot dx$

$= \frac{1}{\ln a}\; a^{f(x)} + c.$

Examples

$$\int \cos x \cdot 2^{\sin x} \cdot dx$$

$$\int \operatorname{cosec} x \cdot \cot x \cdot 8^{\operatorname{cosec} x} \cdot dx$$

$$\int \frac{\cos\sqrt{x} \cdot 3^{(\sin\sqrt{x}+2)}}{\sqrt{x}} \cdot dx .$$

$$\int \sec^2 (3^x + x^2) \cdot (3^x \cdot \ln 3 + 2x) \cdot dx .$$

VII $\int \sinh x \cdot dx = \cosh x + c$

$\int \cosh x \cdot dx = \sinh x + c$

$$\int \sinh(f(x)) \cdot f'(x)\,dx = \cosh(f(x)) + c$$

$$\int \cosh(f(x)) \cdot f'(x)\,dx = \sinh(f(x)) + c.$$

Examples

$$\int e^{\sinh x} \cdot \cosh x \cdot dx$$

$$\int \frac{\sinh(\ln x)}{x} \cdot dx$$

$$\int \cosh(3^x) \cdot 3^x \cdot dx.$$